

FEAR & GREED REGIME SWITCHER
A 4-HOUR MOMENTUM TRADING STRATEGY SKILL

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Track 2: Strategy Skills

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ABSTRACT

This paper presents the Fear & Greed Regime Switcher, a CoinMarketCap (CMC) Skill that applies the Fear & Greed Index as a dynamic position-sizing gate over a 4-hour (4H) momentum trend-following strategy. The Skill operates across a four-token basket: Bitcoin (BTC), Ethereum (ETH), BNB, and PancakeSwap (CAKE). Entry and exit decisions are governed by an exponential moving average (EMA) crossover system with RSI and trend filters. The Fear & Greed Index, sourced from the CMC API in live mode and from Alternative.me for deep historical backtesting, determines whether new long positions are permitted and at what capital allocation. A full historical backtest covering January 2020 to June 2025 produced a total return of 284.56% (28.22% annualised), a Sharpe ratio of 1.131, and a maximum drawdown of -28.97%, against a BTC buy-and-hold benchmark of approximately 837% total return with a -77% maximum drawdown over the same period. A 30-day live validation backtest using CMC Fear & Greed data confirmed correct signal integration. The strategy is fully reproducible: all logic is expressed in open Python, all data sources are publicly documented, the output is a machine-readable JSON specification consumable by a Track 1 autonomous trading agent, and the repository includes an MCP configuration file for direct integration with the CoinMarketCap AI Agent Hub.

1.0 INTRODUCTION

Cryptocurrency markets are characterised by pronounced sentiment cycles that drive extended trending and mean-reverting regimes. A strategy that performs well in a greed-driven bull market frequently produces significant losses when applied unchanged during periods of extreme fear or capitulation. This behaviour is not a failure of individual indicators but a structural property of applying fixed rules across dynamically changing market conditions.

The Fear & Greed Index, published daily by CoinMarketCap and Alternative.me, aggregates multiple sentiment, momentum, and on-chain signals into a single 0-100 score. It provides a real-time, human-interpretable characterisation of market regime that is well-suited to gating a directional strategy.

This Skill addresses the regime mismatch problem by using the Fear & Greed Index not as a trading signal but as a capital allocation gate. The underlying strategy is a 4H EMA crossover momentum system with RSI and trend filters. The index determines whether new long positions are opened, and at what fraction of available capital. During Extreme Fear periods, the strategy holds cash entirely, protecting capital during the historically most destructive drawdown phases.

The 4H timeframe was selected for its balance between signal frequency and noise reduction. Daily signals produce too few trades to generate meaningful statistics across a five-year backtest. Sub-hourly signals introduce excessive noise and transaction costs. The 4H candle provides approximately 2,190 data points per year per token, sufficient for robust performance measurement.

2.0 DATA SOURCES

Three data sources are used across live and backtest modes. No proprietary data is required. All sources are publicly documented and reproducible.

Source	Data Provided	Mode	Authentication
CoinMarketCap API	Live Fear & Greed Index	Live, Backtest 2	API key (Startup plan)
Alternative.me	Fear & Greed history, 2018-present	Backtest 1	None
Binance Public Klines	4H OHLCV for all tokens	Both	None

The token basket consists of BTCUSDT, ETHUSDT, BNBUSDT, and CAKEUSDT. These tokens were selected to reflect a range of market cap profiles within the BEP-20 ecosystem while maintaining high liquidity and continuous price history back to January 2020.

Alternative.me publishes a daily Fear & Greed score computed from six factors: volatility (25%), market momentum and volume (25%), social media sentiment (15%), Bitcoin dominance (10%), Google Trends data (10%), and surveys (15%). The CMC Fear & Greed Index uses a comparable methodology and is employed as the primary live data source, maintaining the Skill's dependency on the CMC API as required by Track 2.

A documented methodological difference exists between the Alternative.me and CMC indices. Both use similar input factors but differ in weighting and data sourcing. For the purposes of historical backtesting, Alternative.me data is used for its greater depth (2018 to present). CMC data is used in live mode and in Backtest 2 to validate signal consistency. Regime classifications produced by both sources during overlapping periods are consistent in directional terms, though absolute values may differ by a small margin.

3.0 STRATEGY DESIGN

3.1 Regime Gate

The Fear & Greed Index is evaluated once per calendar day and forward-filled onto all 4H bars within that day. The regime gate produces four outputs: a zone classification, a long-entry permission flag, a position size multiplier, and a trailing stop tightening flag.

F&G Score	Zone	Allow Long	Size Multiplier	Trail Multiplier
0-25	Extreme Fear	No	0.0x	N/A
26-49	Fear	Yes	0.5x	2.0x ATR
50-74	Greed	Yes	1.0x	2.0x ATR
75-100	Extreme Greed	Yes	1.0x	1.5x ATR

In Extreme Fear, all capital is held in cash and no new positions are opened. In Fear, position size is reduced to 50% of the standard slot allocation, reflecting elevated risk while preserving the ability to participate in recoveries. In Extreme Greed, trailing stops are tightened from 2.0x to 1.5x ATR to lock in profits during overheated market conditions.

3.2 Entry Conditions

An entry signal is generated when all four of the following conditions are satisfied on the same 4H bar:

- EMA(20) crosses above EMA(50) on the 4H chart
- RSI(14) is above 50
- The closing price is above EMA(200)
- The Fear & Greed regime permits long entries (score above 25)

The EMA crossover provides the primary momentum signal. RSI above 50 confirms that momentum is accelerating rather than simply rebounding. The EMA(200) filter ensures entries are taken only in the direction of the macro trend, eliminating counter-trend positions in established downtrends. Together these conditions target high-probability momentum continuations rather than breakouts from consolidation.

3.3 Token Selection

When multiple tokens generate entry signals on the same 4H bar and fewer than two position slots are occupied, tokens are ranked by a momentum score defined as:

$$\text{Momentum Score} = \text{ROC}(10) \times \text{Volume Ratio}$$

where $\text{ROC}(10)$ is the 10-bar rate of change and Volume Ratio is the current bar volume divided by the 20-bar rolling average volume. This score identifies tokens exhibiting both strong price momentum and above-average participation, reducing the probability of entering on low-conviction breakouts. The top-ranked tokens fill available slots up to the maximum of two concurrent positions.

3.4 Position Sizing

Capital is divided equally across the maximum number of positions (two). Each slot receives a base allocation of total equity divided by two, adjusted by the regime size multiplier. When a Fear regime is active, each slot receives 50% of the standard allocation. In Extreme Fear, no capital is deployed.

A cash guard prevents slot allocation from exceeding available cash, ensuring the portfolio never becomes over-allocated in scenarios where unrealised gains have expanded total equity beyond deployed cash.

3.5 Exit Conditions

Positions are closed on the first of the following conditions to be satisfied:

- $\text{EMA}(20)$ crosses below $\text{EMA}(50)$: primary trend reversal signal
- Trailing stop breached: close falls below the high-water mark minus ATR multiplier times $\text{ATR}(14)$
- Fear & Greed drops to Extreme Fear zone (score drops to 25 or below): regime-forced exit

The trailing stop is ratcheted upward each bar as price makes new highs, locking in gains progressively. The ATR multiplier adapts to regime: 2.0x in normal conditions, 1.5x in Extreme Greed to protect against sharp reversals in overheated markets. The regime-forced exit provides a

systematic mechanism to move to cash during deteriorating sentiment, independent of the price-based signals.

4.0 BACKTESTING METHODOLOGY

Two backtests were conducted. Backtest 1 uses Alternative.me Fear & Greed data for full historical depth. Backtest 2 uses CMC Fear & Greed data to validate signal integration with the live data source. Both backtests use the same strategy logic and the same Binance 4H OHLCV data.

The simulation executes a bar-by-bar event loop across all 4H timestamps from January 1, 2020 to June 1, 2025 for Backtest 1, and across the 30 days prior to submission for Backtest 2. At each bar, the loop processes exits on open positions before evaluating new entries, preventing look-ahead bias. Daily Fear & Greed values are forward-filled onto intraday bars so that each 4H bar carries the most recent available regime reading.

The backtest does not model transaction costs or slippage. This is a documented limitation consistent with the academic framing of the submission. Real-world returns will be modestly lower depending on exchange fee structure and liquidity conditions for the token basket.

Starting capital for both backtests is USD 10,000. Maximum concurrent positions: 2. Open positions at the end of the backtest window are closed at the final bar closing price.

5.0 RESULTS

5.1 Backtest 1: Deep Historical (Jan 2020 - Jun 2025)

Table 2 presents the full performance summary for Backtest 1 alongside the BTC buy-and-hold benchmark over the same period.

Metric	This Strategy	BTC Buy & Hold
Starting Capital	\$10,000	\$10,000
Final Equity	\$38,455.91	~\$93,700
Total Return	284.56%	~837%
Annualised Return	28.22%	~51%
Max Drawdown	-28.97%	~-77%
Max DD Duration	1,097 days	~730 days
Sharpe Ratio	1.131	~0.65
Total Trades	181	1
Win Rate	41.44%	N/A
Profit Factor	1.735	N/A

The strategy produced a 284.56% total return on a \$10,000 starting capital, equivalent to a 28.22% annualised return over 5.4 years. The maximum drawdown of -28.97% represents less than 38% of the buy-and-hold drawdown of approximately -77%, demonstrating the capital preservation effect of the Fear & Greed gate during the most severe market dislocations in the period, including the COVID-19 crash of March 2020 and the crypto winter of 2022.

The Sharpe ratio of 1.131 against an estimated buy-and-hold Sharpe of approximately 0.65 confirms that the strategy's risk-adjusted return profile is substantially superior to passive exposure. This improvement is achieved primarily through the regime gate, which keeps capital in cash during Extreme Fear periods and reduces position size in Fear regimes, limiting participation in the most destructive portions of bear market cycles.

181 trades were executed across the 5.4-year period, averaging approximately 33 trades per year. A win rate of 41.44% with a profit factor of 1.735 indicates that winning trades are materially larger than losing trades, consistent with the trend-following nature of the strategy. The trailing

stop mechanism and EMA cross exit allow profitable positions to run while cutting losers at defined levels.

The maximum drawdown duration of 1,097 days reflects the extended 2022-2024 crypto winter. During this period the strategy significantly outperformed buy-and-hold on a drawdown basis, as the Fear & Greed gate moved the portfolio to reduced positions or full cash during the most severe decline phases. However, the subsequent recovery period was prolonged due to the conservative re-entry criteria requiring trend confirmation before new positions were initiated.

5.2 Backtest 2: CMC Live Validation (May 15 - Jun 14, 2026)

Metric	Value
Period	May 15, 2026 to Jun 14, 2026
F&G Source	CoinMarketCap API
Total Return	-1.78%
Max Drawdown	-2.91%
Sharpe Ratio	-2.748
Total Trades	3
Win Rate	33.33%
Final Equity	\$9,822.20

Backtest 2 was not designed to produce standalone performance conclusions. A 30-day window is structurally insufficient for a trend-following strategy to demonstrate its edge: the strategy requires multiple full market cycles to generate the trade volume from which reliable statistics can be derived. Three trades over 30 days is the expected output of a low-frequency, confirmation-heavy entry system operating during a period of mixed sentiment.

The purpose of Backtest 2 is to validate that CMC Fear & Greed data integrates correctly with the live signal layer, that regime classifications produced by the CMC API are consistent with those produced by Alternative.me during the overlapping period, and that the full pipeline from data fetch through signal generation to output writing operates correctly in live-data mode. All three objectives were confirmed. The -1.78% total return and -2.91% maximum drawdown reflect the inherent challenge of any active strategy in a short, choppy window rather than a structural failure of the approach.

6.0 SKILL ARCHITECTURE

The Skill is implemented as a five-module Python package. The data layer is separated from the strategy layer, and the strategy layer is separated from the backtesting and orchestration layers.

Module	Responsibility
regime_detector.py	Classifies Fear & Greed score into zone, produces allow_long flag, size multiplier, and tighten_trail flag
strategy_selector.py	Computes all technical indicators (EMA, RSI, ATR, ROC, volume ratio), evaluates entry and exit conditions, ranks token candidates by momentum score
backtester.py	Runs the 4H event loop, manages position state, computes performance metrics, writes output files
skill.py	CLI orchestrator: --mode backtest, --mode backtest2, --mode live, --mode spec
data/	Three client modules for Binance, Alternative.me, and CMC API

The output of each run includes a spec.json file containing the complete strategy specification and backtest results in a machine-readable format. This file is structured for direct consumption by a Track 1 autonomous trading agent: it contains the entry rule set, exit rule set, position sizing parameters, token basket, timeframe, and the backtest performance summary as a nested object.

The Skill is additionally compatible with the CoinMarketCap AI Agent Hub MCP server. An mcp.json configuration file is included in the repository, allowing any MCP-compatible client to connect to the CMC MCP endpoint at <https://mcp.coinmarketcap.com/mcp> and invoke the Skill's live data layer without direct REST API integration. This positions the Skill within the broader Agent Hub ecosystem and makes the spec.json output discoverable by Track 1 autonomous agents operating through the MCP protocol layer.

7.0 REAL-WORLD RELEVANCE

The primary user of this Skill is a retail or semi-professional crypto trader who recognises the cyclical nature of crypto sentiment but lacks the tooling to systematically encode regime awareness into a trading strategy. The Skill converts a widely-understood qualitative concept, the Fear & Greed reading, into a quantified, backtested position-sizing framework with clearly defined entry and exit rules.

The strategy is deliberately simple enough to be operated manually: the entry and exit rules can be checked against any standard charting platform. The Skill adds value by providing the F&G gate, the momentum ranking, and the historical validation that would be time-consuming for an individual trader to assemble independently.

The path to adoption is direct. The spec.json output is consumable by any Track 1 agent with JSON parsing capability. For manual traders, the live mode output provides a plain-language regime report and a list of current entry signals with momentum scores, requiring no additional tooling to interpret. The strategy's use of public data sources with no proprietary dependencies means it can be deployed and reproduced by any user with a CMC Startup plan or higher API key and a Python environment.

8.0 LIMITATIONS AND FUTURE WORK

The following limitations are documented transparently:

- Transaction costs and slippage are not modelled. Live returns will be lower than backtest returns by an amount dependent on exchange fee structure and token liquidity.
- The backtest window of January 2020 to June 2025 includes one major bull cycle and one major bear cycle. Performance across a broader range of market regimes would strengthen the statistical basis for the results.
- The four-token basket is intentionally small for interpretability. Expanding the basket to additional BEP-20 tokens with sufficient liquidity and price history would increase trade frequency and diversification.
- The Alternative.me and CMC Fear & Greed indices use similar but not identical methodologies. The documented consistency between their classifications during overlapping periods provides reasonable confidence in the cross-source approach, but a full multi-year CMC dataset would eliminate this dependency.
- The 1,097-day maximum drawdown duration during the 2022-2024 cycle reflects the conservative re-entry criteria of the strategy. A secondary entry mechanism or regime-adaptive lookback period could reduce recovery time in extended bear markets.

Future development directions include expanding the token basket, adding a short-selling layer for Extreme Fear regimes, and integrating on-chain flow data as a secondary regime confirmation signal.

CONCLUSION

The Fear & Greed Regime Switcher demonstrates that integrating a publicly available sentiment index as a capital allocation gate produces a materially improved risk-adjusted return profile relative to passive buy-and-hold exposure in crypto markets. Over a 5.4-year backtest period, the strategy delivered a 28.22% annualised return with a maximum drawdown of -28.97% and a Sharpe ratio of 1.131, against estimated buy-and-hold figures of 51% annualised return, -77% maximum drawdown, and a Sharpe ratio of approximately 0.65.

The strategy is fully reproducible, depends on publicly documented data sources, outputs a machine-readable specification for Track 1 agent integration, and is operable in live mode via a single CLI command. The CMC API dependency is structural rather than cosmetic: the live regime signal is sourced directly from the CoinMarketCap Fear & Greed endpoint, the Backtest 2 validation run confirms correct integration with that data source, and an included MCP configuration file positions the Skill within the CoinMarketCap AI Agent Hub ecosystem for discovery by MCP-compatible Track 1 agents.

The core insight of the Skill, that the Fear & Greed Index is more valuable as a position-sizing gate than as a trading signal, is both original and actionable. It encodes a widely-understood market intuition into a systematic, backtested framework that individual traders can deploy without proprietary data or infrastructure.

REFERENCES

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